

# **Three Operations from $T^4$ to Observable Space**

**Demarcation from Discrete Emergence Models, with Case Study  
HLV**

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### Correction notes

**Corrections from the calculation review (1 Oct. 2026).** The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Flat torus: Gaussian curvature is intrinsic; on the ring torus in  $\mathbb{R}^3$  it is pointwise  $\neq 0$ , only its integral vanishes (previously “sums to zero pointwise”).
- Fractal operator:  $r_n = \xi^n \rightarrow r_n \in (0, 1)$ , concrete choice open (with  $r_n = \xi^n$  one would have  $\|\hat{F}\| \leq 1.00013 \xi$ ); the particle masses are eigenvalues of the mass operator  $\hat{M}$ , not of  $\hat{F}$ ;  $\hat{F}$  is self-adjoint on the  $\mathbb{Z}_3$ -invariant sector (Doc. 327, Theorem 3).

## Abstract

This document consolidates the demarcation of FFGFT against discrete emergence models (here: HLV by Krüger, August 2026) and clarifies which operations mediate between  $T^4$  and the observable space  $\mathbb{R}^3 \times \mathbb{R}_t$ .

Central claim: The failure of discrete lattice models (class-X4 overlaps, missing operator specificity, open continuum limit) is not a surprise, but a structural consequence of treating geometric auxiliary constructs as ontological building blocks. FFGFT avoids these failures not by choosing a different lattice, but because the fundamental level  $T^4$  is compact and geometrically complete — and the three operations leading from it to the observable space have strictly different logical status.

The three operations (Doc. 270 **[K]**):

1. **Type I** — Duality decompactification:  $S_m^1 \rightarrow \mathbb{R}_t$  via  $\tilde{T} \cdot m = 1$ . Information-preserving (embedding, not projection).
2. **Type II** — Geometric projection:  $D_3 \rightarrow D_2 \rightarrow D_1$ . Lossy, irreversible. The step  $D_3 \rightarrow D_2$  is identified as holographic projection.
3. **Type III** — Representational translation:  $T^4 \leftrightarrow \mathcal{H}$  (Hilbert space/matrix). Bijective, lossless — upward movement, no geometric dimension reduction.

## .1 Starting point: the HLV audit as case study

Marcel Krüger subjected his own Helix-Light-Vortex model (HLV) to four prospectively frozen mathematical tests in *Local-to-Global Compatibility on Aperiodic Cut-and-Project Carriers* (v1.4, August 2026). All key claims fail or remain open.

1. **Local-to-global compatibility:** 5,455 genuine volume overlaps (class X4) among 3,384 projected tetrahedra. Six witnesses certified with 120-digit arithmetic. *Failure.* **[E]**
2. **Operator specificity:** An ordinary periodic cubic lattice (CUBIC) outperforms HLV in spectral multiplet score (GAE-001A). *Rejected.* **[E]**
3. **Geometry-to-constitutive law:** Pure carrier geometry does not determine physical densities, masses, or couplings (Proposition 2.2). *Trivially negative.* **[E]**
4. **Finite-to-continuum:** A bounded graph has bounded spectrum, no UV limit. Mosco/resolvent convergence unproven. *Open.* **[E]**

### Corollary (Krüger 4.5) **[E]**

Local validity of an operator does not imply global embeddability of the underlying carrier.

## The structural cause

HLV attempts to use geometric bodies (3D tetrahedra from  $\mathbb{R}^6$  projection) as ontological building blocks of physical space. It thereby inherits all four problems. The remedy is not a *better* lattice — it is a fundamental carrier that is geometrically complete and closed from the outset.

## .2 The FFGFT fundamental level: $T^4$ with D4 lattice

### Structure of $T^4$

The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S_x^1 \times S_y^1 \times S_z^1 \times S_m^1$$

with three spatial circles  $S_{x,y,z}^1$  (radius  $R_H$ , quasi-continuous) and one mass circle  $S_m^1$  (radius  $R_m \sim \xi \cdot \ell_P$ , discrete). The fourth direction  $S_m^1$  is not a mass dimension alongside a separate time, but the substitute-time direction itself. Mass and time are two readings of *the same* dimension;  $\tilde{T} \cdot m = 1$  is the translation dictionary (Doc. 270 [K], Doc. 314 [K]). **[POSTULATE]**

The lattice on  $T^4$  is the D4 lattice (Doc. 314 [K]): kissing number 24,  $|\text{Aut}(D_4)| = 1152$ , triality as built-in  $\mathbb{Z}_3$  symmetry (16 order-3 elements generate the  $T^4/\mathbb{Z}_3$  orbifold with 9 fixed points). Negative result with theorem character:  $5 \nmid 1152$ , so  $\text{Aut}(D_4)$  contains no order-5 element [K].

### Why $T^4$ is flat — intrinsic vs. extrinsic curvature [K]

A ring torus embedded in  $\mathbb{R}^3$  ( $T^2 \hookrightarrow \mathbb{R}^3$ ) is curved: the outer side is convex (positive Gaussian curvature), the inner side concave (negative Gaussian curvature). By the Theorema Egregium this Gaussian curvature is *intrinsic*, an isometric invariant; on the ring torus it is pointwise non-zero,  $K = \cos v / (r(R + r \cos v))$ , and only its integral vanishes ( $\int K dA = 2\pi\chi = 0$ , Gauss–Bonnet). Intrinsically flat ( $K \equiv 0$ ) is the flat torus  $\mathbb{R}^n/\Lambda$ , which cannot be embedded isometrically in  $\mathbb{R}^3$ ;  $T^4 = \mathbb{R}^4/\Lambda$  is of this kind. (Corrected on 1 Oct. 2026: previously “The intrinsic Gaussian curvature, however, sums to zero pointwise” –  $K$  follows from the metric alone, is  $> 0$  outside,  $< 0$  inside,  $\int K dA = 0$ ; the flat torus has  $K = 0$ .)

#### Intrinsically flat torus [K]

$T^4$  is intrinsically flat: an interior observer measures no curvature. Angle sums are  $\pi$ , parallel geodesics remain parallel.  $T^4$  is locally isometric to  $\mathbb{R}^4$  — with global boundary identification, but without local curvature corrections.

Two direct consequences for FFGFT:

**(a) D4 lattice through topology, not curvature.** What the D4 lattice carries into the Hilbert space is topological data: winding numbers, mode content, triality. The distinction  $\mathbb{Z}^4/D_4$  is spectral, not geometric (Doc. 314 [K]: “The torus is flat anyway — curvature was never the difference”).

**(b) Fourier analysis without curvature corrections.** On the flat torus, Fourier modes  $e^{ik \cdot x}$  form the natural basis; Laplacian eigenvalues are  $|k|^2$  without curvature correction. This is the basis for the bijective, lossless Type III translation  $T^4 \leftrightarrow L^2(T^4) \otimes \mathbb{C}^3$ : Fourier decomposition on a flat torus is a complete isometric map.

This structurally distinguishes FFGFT from approaches like HLV, which embed geometric bodies into  $\mathbb{R}^3$  and thereby generate overlap problems: the embedding question does not arise in FFGFT, because  $T^4$  as an intrinsically flat, self-contained carrier requires no embedding into an outer space.

## How information is encoded in the D4 lattice [K]

The geometry does not disappear when moving to the Hilbert space — it migrates into spectrum and fibre (Doc. 314 [K]):

**Spectrum = theta series.** The degeneracies of the Laplacian on flat  $T^4$  are exactly the theta series of the D4 lattice:  $\theta_{D_4}(q) = 1 + 24q^2 + 24q^4 + \dots$ . The kissing number 24 becomes the degeneracy of the first excitation  $|k|^2 = 2$ . Two flat tori ( $\mathbb{Z}^4$  vs.  $D_4$ ) are distinguishable by spectrum alone, without curvature [K].

**Fibre = triality.** Triality is built in, not appended [K]:  $|\text{Aut}(D_4)| = 1152$ , quotient to  $W(D_4)$  is  $6 = |S_3|$ . The 16 order-3 elements generate the  $\mathbb{Z}_3$  orbifold and act on every mode orbit as a  $\mathbb{C}^3$  circulant. The fibre  $\mathbb{C}^3$  follows from the lattice automorphism structure, not an additional postulate.

**Winding numbers = topology.** The reciprocity relation  $\lambda_4 \cdot m = 2\pi$  holds mode-by-mode exactly [K]: winding number = mass number. Winding numbers are topological — invariant under deformation; fractal corrections accumulate only when unrolled, not when rolled up.

In one sentence (Doc. 314): “In the Hilbert space the lattice becomes flat, but not without consequence — what geometry carried, spectrum and fibre carry, and the division of labour between them is exactly determinable.”

## Where $\tilde{T} \cdot m = 1$ appears in the Hilbert space [K]

The time-mass duality appears in  $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$  not as time evolution, but in three complementary forms:

**(a) As the fourth winding number  $n_4$ .** The multi-index  $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$  counts windings in all four torus directions [B] (Doc. 322). The mass dimension  $S_m^1$  carries the winding number  $n_4$ . The reciprocity relation  $\lambda_4 \cdot m = 2\pi$  holds mode-by-mode exactly [K] (Doc. 314): winding number = mass number.  $\tilde{T} \cdot m = 1$  is thus the Kaluza-Klein relation of the fourth torus direction, readable as  $n_4$  quantum number in the spectrum.

**(b) As the mass operator eigenvalue equation.**

$$\hat{M} |n_\theta, n_\phi, p\rangle = m_i |n_\theta, n_\phi, p\rangle \quad [\text{K}] \quad (.1)$$

The eigenvalues are the particle masses (Doc. 322, Doc. 320). The duality appears as the spectrum of the operator.

**(c) As radius-class separation in the spectrum.** The splitting at  $R_4 \neq R_{123}$  makes  $\tilde{T} \cdot m = 1$  directly readable [K] (Doc. 314): the mass dimension generates a large mode spacing  $\sim 1/R_m$ , the spatial directions a quasi-continuous spacing  $\sim 1/R_H$ .

### What does *not* appear in the Hilbert space

Time  $\mathbb{R}_t$  itself. It is the result of the Type-I decompactification  $S_m^1 \rightarrow \mathbb{R}_t$  — a geometric downward operation *before* the Type-III translation. In the Hilbert space  $L^2(T^4)$ , only the torus lives; time appears only when leaving the Hilbert space and embedding the spectrum into  $\mathbb{R}_t$ .  $\tilde{T} \cdot m = 1$  resides in the Hilbert space as  $n_4$  winding number and mass operator eigenvalue structure, not as time evolution.

## Where the fractal property resides in the Hilbert space [K] / [B]

The fractal property of FFGFT appears in the Hilbert space at three precisely separable locations:

**(a) In the operator  $\hat{F}$  itself — as self-similarity structure [B].**

$$\hat{F} = \bigoplus_{n=1}^N S_n \circ L_n, \quad S_n = r_n U_n, \quad r_n \in (0, 1) \quad (.2)$$

(Corrected on 1 Oct. 2026: previously  $r_n = \xi^n \in (0, 1)$  – Doc. 322 has  $r_n \in (0, 1)$ , the concrete choice is open. With  $r_n = \xi^n$  one would have  $\|\hat{F}\| \leq \sum_n \xi^n = \xi/(1 - \xi) = 1.33351 \cdot 10^{-4} = 1.00013 \xi$ ; together with  $\xi = \lambda_{\min}$ ,  $\xi$  would also be an upper spectral bound, and the  $\hat{F}$  spectrum would leave no room for a mass hierarchy.) The scale operators  $S_n$  implement discrete scale transformations; each level  $n$  is a contraction of the previous by  $r_n$  — this is self-similarity as an operator.  $\hat{F}$  encodes the sub-Planck geometry as a hierarchical convolution of scale levels down to the minimum length scale  $L_0 = \xi \cdot \ell_P$  (Doc. 180, 327 [K]).

**(b) In the fractal dimension  $D_f = 3 - \xi$  — as spectral property [K].**

$$D_f = \inf \left\{ s : \sum_i \lambda_i^s < \infty \right\} = 3 - \xi \approx 2.9999 \quad [\mathbf{K}] \quad (.3)$$

$D_f$  is defined via the Hausdorff measure of the spectrum of  $\hat{F}$  (Doc. 322): the spectrum on  $L^2(T^4)$  has a fractal fine structure. The value is spatial ( $3 - \xi$ , three dimensions), not four-dimensional. Local action:  $D_f^{\text{space}} = 3 - \xi$  with  $6.7 \times 10^{-5}$  relative (Doc. 314 [K]).

**(c) In the interface factor  $K_{\text{frac}}$  — as scale bridge [K].**

$$K_{\text{frac}} = 1 - 100\xi = \frac{74}{75} \approx 0.9867 \quad (.4)$$

$K_{\text{frac}}$  is cumulative (Doc. 133): it accumulates over the RG running across 17 orders of magnitude and acts at the transition from geometric (bare) to measured (SI) mass:  $m_i^{\text{SI}} = K_{\text{frac}} \cdot C_{\text{conv}} \cdot m_i^{\text{bare}}$ .

### Decisive separation [K]

In the *rolled-up*  $T^4$  (Hilbert space) there are **no** fractal corrections — winding numbers are purely topological (Doc. 314 [K]). The fractal property appears only when unrolling (Type I:  $S_m^1 \rightarrow \mathbb{R}_t$ ) as an accumulated defect. In the Hilbert space itself,  $\hat{F}$  is a bounded operator, self-adjoint with real spectrum on the  $\mathbb{Z}_3$ -invariant sector [B] (Doc. 327, Theorem 3) — the fractality resides in the scale hierarchy of the operator structure ( $r_n \in (0, 1)$ ), not in the carrier  $L^2(T^4)$ . (Corrected on 1 Oct. 2026: previously “bounded self-adjoint operator ... ( $r_n = \xi^n$ )” – see the note under (a) above and Doc. 327, Theorem 3.)

### .3 No embedding problem in $T^4$

$T^4$  is compact, complete, and geometrically self-contained. There is no embedding problem because  $T^4$  does not need to be *embedded* in an outer space: it *is* the fundamental level. The class-X4 errors of HLV arise because projection elements must be embedded into an external  $\mathbb{R}^3$ . In FFGFT this question does not arise.

#### Type III: representational translation upward

The translation  $T^4 \leftrightarrow \mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$  (Doc. 230, 314, 322) is a bijective, lossless upward movement (Type III, Doc. 270 **[K]**). It adds no geometric extra dimensions, but expresses the same  $T^4$  content in a higher-dimensional language. The mass operator  $\hat{M}$  (Doc. 322, section on masses) has the particle masses as eigenvalues;  $\xi = \lambda_{\min}(\hat{F}_{D_4})$  is the smallest eigenvalue of the  $D_4$  sub-operator of  $\hat{F}$  **[B]**, provided the  $D_4$  sector lies in the  $\mathbb{Z}_3$ -invariant sector, on which  $\hat{F}$  is self-adjoint (Doc. 327, Theorem 3 and Corollary 7). *(Corrected on 1 Oct. 2026: previously “The spectrum of  $\hat{F}$  (Doc. 322, self-adjoint **[B]**, Doc. 327) has the particle masses as eigenvalues” – with  $\xi = \lambda_{\min}$  and  $\|\hat{F}\| \leq \xi/(1-\xi)$  the  $\hat{F}$  spectrum would leave no room for  $m_\tau/m_e = 3477$ ; in Doc. 322 the masses are eigenvalues of  $\hat{M}$ .)*

### .4 The three operations from $T^4$ to the observable space

#### Type I: duality decompactification

$$S_m^1 \xrightarrow{\tilde{T} \cdot m=1} \mathbb{R}_t$$

The unrolling of the compact mass dimension into the non-compact time direction is an *embedding*, not a projection. No mode content is lost; time arises as an additional non-compact direction **[K]**.

#### Type II: geometric projection (lossy)

The further steps  $D3 \rightarrow D2 \rightarrow D1$  are genuine geometric projections of spatial directions: lossy, irreversible. The step  $D3 \rightarrow D2$  is identified as holographic projection (Doc. 270 **[K]**). It is not fundamental.

##### Classification of the holographic step

The step  $D3 \rightarrow D2$  is *not* the fundamental level of FFGFT. It is a lossy observer projection — physically relevant for surface phenomena (black hole membrane, Doc. 325), but not for the derivation of masses and couplings anchored on  $T^4$ . The two-dimensional projection surface is not a fundamental object.

| Type | Operation                                                      | Direction             | Status                            |
|------|----------------------------------------------------------------|-----------------------|-----------------------------------|
| I    | Duality decompactification $S_m^1 \rightarrow \mathbb{R}_t$    | geometrically down    | information-preserving <b>[K]</b> |
| II   | Geometric projection $D3 \rightarrow D2 \rightarrow D1$        | geometrically down    | lossy <b>[K]</b>                  |
| III  | Representational translation $T^4 \leftrightarrow \mathcal{H}$ | representationally up | bijective, lossless <b>[K]</b>    |

## Three-way comparison

## .5 Open questions in the flat Hilbert space — status

### Question 1: Haar measure and D4 lattice **[B]**

The canonical Haar measure on  $T^4$  is compatible with the D4 spectral structure (scripts 330-1, all 7/7): Fourier modes are exactly orthogonal; D4 contains only even norm shells;  $\text{Aut}(D_4) \subset \text{Iso}(T^4)$  leaves the Haar measure invariant. The  $\mathbb{Z}_3$  orbifold breaks translation invariance and splits the 24-fold degeneracy to  $9 \oplus 8 \oplus 4 \oplus 2 \oplus 1$  (Doc. 314 **[K]**) — but only as fine structure  $\sim \xi \approx 10^{-4}$ , below any measurement threshold. **[B]**

### Question 2: Orbifold fixed points **[K]** / **[S]**

Nine fixed points of  $T^4/\mathbb{Z}_3$  follow from the Lefschetz fixed-point formula (script 330-2, B1):  $|\det(1 - A)| = |(1 - \omega)(1 - \omega^2)|^2 = 9$ . Neutrino localisation:  $\nu_1 \leftrightarrow F_2$ ,  $\nu_2 \leftrightarrow F_5$ ,  $\nu_3 \leftrightarrow F_7$  **[K]** (Doc. 322). What remains **[S]**: back-reaction of Hawking emission on the fixed-point structure during  $M \rightarrow M_{\text{coll}} = m_P/\sqrt{2}$  (explicitly open, Doc. 325).

### Question 3: Fractal weighting $r_n$ and $L^2(T^4)$ **[B]**

$r_n \in (0, 1)$  (Doc. 322; concrete choice open) is position-independent:  $\hat{F}$  does *not* break the translation invariance of  $L^2(T^4)$  — only the  $\mathbb{Z}_3$  orbifold does (and only as fine structure  $\sim \xi$ ).  $\hat{F}$  is bounded ( $\|\hat{F}\| \leq \sum r_n < \infty$ ); it is normal and self-adjoint on the  $\mathbb{Z}_3$ -invariant sector of  $\mathcal{H}$  **[B]**, on the whole space open **[S]** (Doc. 327, Theorem 3). (Corrected on 1 Oct. 2026: previously “ $r_n = \xi^n$ ” and “bounded ... and self-adjoint on  $\mathcal{H}$ ” — with  $r_n = \xi^n$ ,  $\|\hat{F}\| \leq 1.00013 \xi$  and  $\xi$  would be an upper spectral bound; boundedness alone does not give self-adjointness.)

#### Status of the three questions

Question 1 (Haar/D4): **[B]** — fully resolved.

Question 2 (fixed points): **[K]** for 9 fixed points and  $\nu$ -localisation; **[S]** for back-reaction during evaporation (explicitly open, Doc. 325).

Question 3 ( $r_n/\hat{F}$ ): **[B]** —  $\hat{F}$  bounded, self-adjoint on the  $\mathbb{Z}_3$ -invariant sector (whole space **[S]**);  $r_n$  does not break translation invariance.

## .6 Why Krüger's four problems do not arise in FFGFT

1. **Class-X4 overlaps:** arise because HLV embeds projection elements into  $\mathbb{R}^3$ . In FFGFT,  $T^4$  is the fundamental level; no embedding into an outer space occurs.
2. **Missing operator specificity:** In FFGFT the spectrum is determined by the D4 lattice with built-in triality. The negative result  $5 \nmid |\text{Aut}(D_4)|$  is itself a specific theorem about the spectrum **[K]**.
3. **Geometry without physics:**  $T^4$  alone also does not fix coupling constants. The physics comes from the combination  $T^4$ -spectrum +  $\xi = \lambda_{\min}(\hat{F}_{D_4})$ .
4. **Continuum limit:** No limit problem, because Type I is an embedding (no limit passage needed) and Type III is bijective (no information loss). Only Type II is lossy — but this is known and documented (Doc. 270).

## .7 What remains from the comparative discussion

**Structurally secured:** The four HLV problems do not arise in FFGFT — due to  $T^4$  as intrinsically flat, self-contained carrier, bijective Type-III translation, and Type-I embedding  $S_m^1 \hookrightarrow \mathbb{R}_t$ .

**Correct from the comparative discussion:** The holographic projection  $D3 \rightarrow D2$  (Type II) is lossy and not fundamental; mass and time are not encoded in the 2D surface but arise through Type I ( $\tilde{T} \cdot m = 1$ ) and the recursion  $\xi^n$ .

**Correction vs. uncritical adoption:** There is no “two-dimensional projection surface as fundamental object” in FFGFT. The corpus has a clear ontological hierarchy:  $T^4$  as primary reality (Doc. 152 **[POSTULATE]**), not a 2D surface.  $\tilde{T} \cdot m = 1$  is the Kaluza-Klein relation of the mass dimension, not a “reading principle on a screen”.

**Verification**                      **scripts:**                      pruef\_330\_1\_haar\_d4.py                      (7/7),  
pruef\_330\_2\_fixpunkte\_rn.py (9/9).

## Status overview

| Element                                                                      | Status             |
|------------------------------------------------------------------------------|--------------------|
| $T^4 = S_x^1 \times S_y^1 \times S_z^1 \times S_m^1$ as fundamental level    | <b>[POSTULATE]</b> |
| D4 lattice on $T^4$ ; kissing number 24; $ \text{Aut}(D_4)  = 1152$          | <b>[K]</b>         |
| $T^4$ intrinsically flat; spectral structure from topology                   | <b>[K]</b>         |
| Information in spectrum (theta series) and fibre (triality)                  | <b>[K]</b>         |
| $\tilde{T} \cdot m = 1$ as $n_4$ winding number and mass operator eigenvalue | <b>[K]</b>         |
| Fractal property in $\hat{F}$ structure ( $r_n = \xi^n$ ), not in $L^2(T^4)$ | <b>[B]</b>         |
| $D_f = 3 - \xi$ ; no fractal corrections in rolled-up $T^4$                  | <b>[K]</b>         |
| Three operation types (Doc. 270): Type I/II/III                              | <b>[K]</b>         |
| Type III bijective ( $T^4 \leftrightarrow \mathcal{H}$ , Doc. 322/314)       | <b>[K]</b>         |
| HLV class-X4 errors structurally excluded (no embedding problem)             | <b>[B]</b>         |
| Continuum limit structurally resolved by Type I (embedding)                  | <b>[K]</b>         |

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